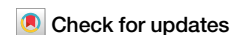


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Identical particles as a genuine non-local resource

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All particles of the same type are indistinguishable, according to a fundamental quantum principle. This entails a description of many-particle states using symmetrised or anti-symmetrised wave functions, which turn out to be formally entangled. However, the measurement of individual particles is hampered by a mode description in the second-quantised theory that masks this entanglement. Is it nonetheless possible to use such states as a resource in Bell-type experiments? More specifically, *which states of identical particles can demonstrate non-local correlations in passive linear optical setups that are conventionally taken to be a classical component of the experiment?* Here, the problem is fully solved for multi-particle states with a definite number of identical particles. We show that *all* fermion states and *most* boson states provide a sufficient quantum resource to exhibit non-locality in passive linear optics. The only exception is a special class of boson states that are reducible to a single mode, which turns out to be locally simulable for any passive linear optical experiment. This finding hints at an intimate connection between the fundamental principle of particle indistinguishability and Bell non-locality, which turns out to be observable with very modest optical means for almost every state of identical particles.

Bell nonlocality stands out as a hallmark of quantum theory, defying our classical conception of reality^{1–4}. It boils down to strange correlations between outcomes of space-time separated measurements that cannot be explained in the usual causal manner implied by the experimental design, i.e. only as an effect of preparation at the source. Quantum theory predicts the appearance of such correlations when the system is prepared in an entangled state, which has been confirmed in a number of ingenious experiments^{5–9}. Inherent to the notion of entanglement is the concept of a composite system described as the tensor product of subsystems, which presupposes that a full range of measurements on individual subsystems can be made^{10,11}. This raises the subtle question of whether a given quantum state can actually demonstrate Bell non-locality. The answer will crucially depend on how the subsystems are defined and which measurements are allowed in the experiment.

The issue is especially relevant in the context of identical particles. According to the fundamental principle of indistinguishability, multi-particle systems are described by entangled states due to the symmetrisation or anti-symmetrisation of the wave function^{12,13}. However, in the second-quantised theory, this fact is obscured by moving to the mode description of the system, where individual particles cannot be directly addressed, thereby preventing explicit probing of subsystems where this entanglement resides.

A sensible criterion of entanglement for identical particles is therefore challenging^{14–22}, which suggests a shift in focus towards operational notions of nonclassicality. One such important notion concerns whether and how states of identical particles can lead to non-local correlations that can be experimentally observed. This allows us to deliberately bypass the question of *entanglement* for a system of identical particles and instead ask directly about *non-local correlations*. Notably, the latter pertains to experimentally observed correlations rather than being a mathematically driven construct.

In this work, we treat passive linear optics (comprised of mirrors, phase shifters, beam splitters, and detectors) as a purely classical component of the experiment. From this perspective, any non-classical behaviour observed must have its origin in the remaining part of the experiment, and thus be attributed to the state that is fed into the classical optical setup. See Fig. 1 for illustration. This view provides a straightforward operational way to assess non-classical potential of a given quantum state with respect to passive linear optics.

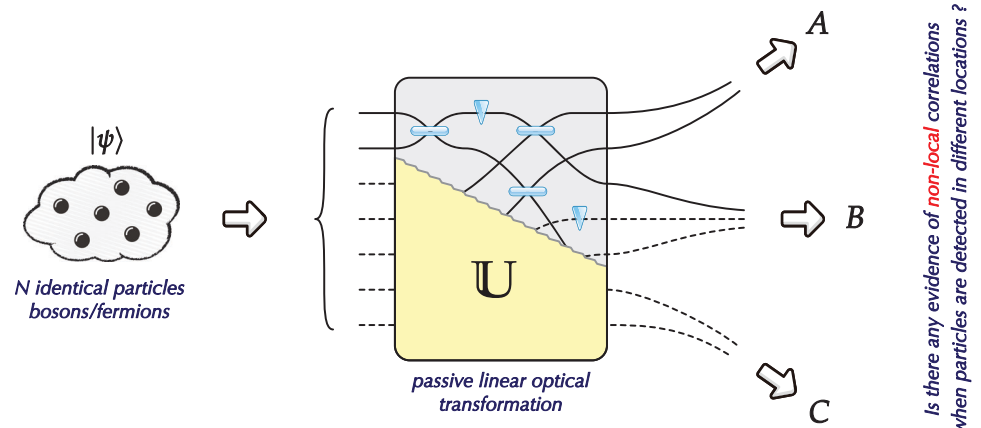
Our research problem has a simple expression in the following question. For a given quantum state of identical particles, we ask:

Does there exist some passive linear optical experiment capable of demonstrating Bell non-locality?

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Fig. 1 | Passive linear optical experiment.

A quantum state $|\psi\rangle$ of N identical particles (bosons or fermions) enters a classical optical setup built of paths, mirrors, phase shifters, beam splitters, and detectors. Arranged in various configurations, optical elements implement a unitary transformation $\mathbb{U}$ on each particle²⁷. Then, the modes/paths are distributed to different locations $A, B, C, \dots$, where the particles can be further processed and finally detected. Each experiment has the potential to reveal non-local correlations in the observed statistics. Here, the state $|\psi\rangle$ processed with the linear optical setup $\mathbb{U}$ is interpreted as a source of particles for a Bell test that is performed at distant locations $A, B, C, \dots$.



A positive answer indicates that the state is a genuine *non-local resource*. Conversely, the lack of any such experiment means that the correlations observed in every optical setup can be explained by a generic local hidden variable model. Then, the state *fails* to provide adequate non-local resources. For example, it was shown that every single-particle state, for both bosons and fermions, can be locally simulated in a generic way²³.

Note that the essence of such a formulated problem lies in the arbitrary nature of passive linear optical experiments, which can be composed in any possible manner. Furthermore, our query covers every single state within a sufficiently broad class of interest. In this article, the problem is fully solved for multi-particle states with a *definite number* of identical particles for both boson and fermion statistics.

We show that all fermion states and the vast majority of boson states provide a genuine non-local resource (with respect to passive linear optics), except for a narrow class of states that are reducible to a single mode.

See Fig. 2 for illustration. This result hints at an intriguing link between the basic concept of indistinguishability of particles and the puzzling feature of Bell non-locality within the framework of classical optical setups.

Results

Informal statement of the problem

Suppose we lived in a purely classical world and someone presented us with a quantum state of N identical particles (either fermions or bosons). *Would we be able to recognise this new quality with our classical tools?* By classical tools, we mean standard passive linear optics, which includes mirrors, phase shifters, beam splitters, and detectors that can be arranged in various configurations. Arguably, these elements are found in any classical optical laboratory, and electromagnetic waves do not exhibit non-classical behaviour in experiments of this type^{24,25}. What can be imagined is that such a quantum state is fed into an optical circuit, processed in any conceivable way, and then the particles are detected in different locations. See Fig. 1. The pattern of correlations observed in such an experiment will be deemed non-classical only if they *cannot* be explained by a local model with some ‘stuff’ (particles, fields and/or hidden variables) propagating along the paths as specified by the experimental design. Then, the state can be called a genuine *non-local resource*. Otherwise, there is an account of every optical experiment that does not require any non-local mechanisms, and therefore the state should not give rise to surprise.

The problem is formulated by thinking of the quantum state as a potential *non-classical resource* with respect to a fixed set of operations (i.e. passive linear optics and detectors). Specifically, we are interested in the following question:

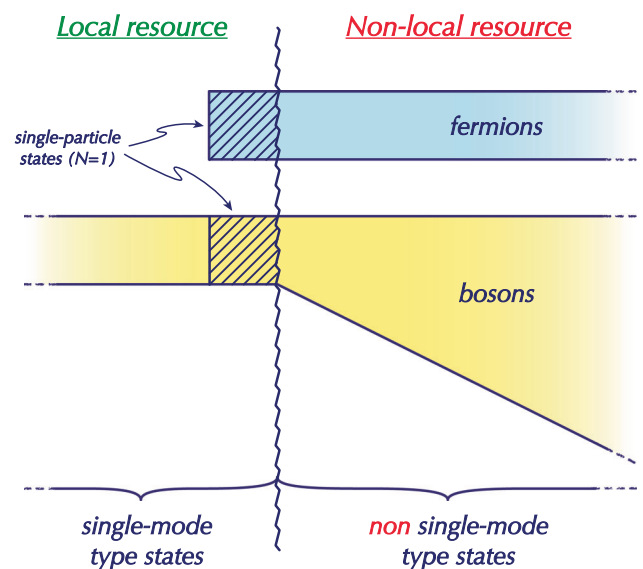


Fig. 2 | Classification of states as a non-local resource (with respect to passive linear optics). An illustration of fermion and boson states in Eq. (1)/(2). The extension of the bands depicts a wider range of boson states than fermion states (due to the Pauli exclusion principle), and the vertical wiggling line separates the single- and non single-mode type states described in **Definition 1**. Our main result **Theorem 1** shows that, for states with a *definite number* of identical particles, the distinction between the states providing *local* vs *non-local* resource (with respect to passive linear optics) corresponds to a division into *single-mode type* vs *non single-mode type* states. More specifically, all fermion states provide a non-local resource, except for single-particle states ($N = 1$). While in the richer family of boson states, the only ones which are incapable of demonstrating non-locality are states reducible to a single mode.

Which quantum states of identical particles can lead to non-local behaviour in classical experiments composed of passive linear optical transformations and detectors?

For each particular state, it has two possible answers:

Yes: The state is a genuine *non-local resource*. It means that the state can be used in some experimental procedure to demonstrate non-local correlations in space-time separated regions (by violating some Bell-type inequality in some specific experiment).

No: The state does *not* provide a genuine *non-local resource*. That is, the correlations observed in any conceivable experiment can be explained

in a local manner (via some generic model in which hidden variables propagate locally, following the pattern of connections in the experimental design).

This splits the space of quantum states into two disjoint classes. The first class can fuel classical optical setups to operate at a non-local level. The second class appears to be classical-like, i.e. simulable with local resources, and hence one should not expect a non-classical advantage in passive linear optics with those states.

Here, we are interested in the classification of all quantum states with a *definite* number of particles (N is fixed). Notably, for a single particle ($N = 1$), the question was resolved in ref. 23, where a generic local hidden variable model of arbitrary passive linear optical experiments was constructed (i.e. single-particle states do not provide a non-local resource). However, a far more complex and interesting problem is that of multi-particle systems ($N \geq 2$). We show that *all* fermion states and *most* boson states provide a genuine *non-local resource*.

Passive linear optical toolkit

In the *occupation number representation* (Fock space), a general state of N identical particles is described by

$$|\psi\rangle = \sum_{n_1 + \dots + n_M = N} \psi_{n_1 \dots n_M} |n_1 \dots n_M\rangle, \quad (1)$$

which is as a (complex) combination of states $|n_1 \dots n_M\rangle$ with a definite number of particles n_i in the respective modes $i = 1, \dots, M$. Those numbers are constrained by the particle statistics: for *bosons* $n_i \in \mathbb{N}$, and for *fermions* $n_i = 0, 1$. In the following, we will consider situations where the total number of particles N is *fixed* and *without* any restriction on the number of modes M . Furthermore, since we are concerned with locality, we will focus on optical modes of the system, i.e. think of each mode as a different path along which particles can propagate.

It will be convenient to use the standard representation in terms of the *creation/annihilation* operators, which add/subtract a particle in the respective mode of the system. Then Eq. (1) can be written in the form

$$|\psi\rangle = \sum_{n_1 + \dots + n_M = N} \psi_{n_1 \dots n_M} \prod_{i=1}^M \frac{a_i^{\dagger n_i}}{\sqrt{n_i!}} |0\rangle, \quad (2)$$

where $|0\rangle$ is the *vacuum* state and the *creation* and *annihilation* operators, $a_i^\dagger$ and a_i , satisfy the usual commutation relations specific for the respective *boson* or *fermion* statistics^{12,13,26}.

In the following, we will be interested in transformations of the Fock states that can be implemented with passive linear optics, i.e. a sequence of single-mode and two-mode operations (or gates) that are conventionally realised by mirrors, phase shifters and beam splitters. Such transformations are described as follows

$$a_i^\dagger \xrightarrow{\mathbb{U}} a_i^\dagger = \sum_{j=1}^M U_{ij} a_j^\dagger, \quad (3)$$

for each mode $i = 1, \dots, M$ of the system, and $\mathbb{U} \equiv [U_{ij}]$ is a unitary transformation on $\mathbb{C}^M$. Furthermore, it is known that any unitary $\mathbb{U}$ on a single-particle space $\mathbb{C}^M$ can be realised by passive linear optics (i.e. can be obtained as a sequence of single- and two-mode operations implemented by mirrors, phase shifters and beam splitters)²⁷. However, we note that this is not enough to generate any transformation on the multi-particle Fock space since under passive linear optics, the state space splits into a myriad of non-equivalent continuous classes whose full classification is beyond reach, see ref. 28.

For further convenience, let us formally distinguish a special class of states that will play an important role in the following discussion.

Definition 1. A state is said to be of a single-mode type if it can be reduced to a single mode by passive linear optics. That is, for a state in Eq. (1)/(2), it means that there exists a unitary $\mathbb{U}$ on $\mathbb{C}^M$ (or a sequence of optical elements) such that

$$|\psi\rangle = \mathbb{U} |N, 0, \dots, 0\rangle = \frac{a_1^{\dagger N}}{\sqrt{N!}} |0\rangle. \quad (4)$$

Equivalently, those states can be thought of as having evolved from the initial state in which all the particles were in the same mode. A straightforward consequence of ref. 27 is that all single-particle states ($N = 1$) are of a single-mode type (for both bosons and fermions). Clearly, in the multi-particle case ($N \geq 2$), the class of single-mode type states is nontrivial only for bosons (it is empty for fermions due to the Pauli exclusion principle). We hasten to note that **Definition 1** boils down to a few easy-to-check conditions; see **Supplementary Information** for a discussion.

Bell non-locality in optical experiments

Every experiment takes place in space and time. This is particularly manifest in optical setups that are composed of basic elements distributed in different positions and interconnected by optical paths into a larger circuit. In a typical optical experiment, an initial state $|\psi\rangle$ is fed into a circuit that is responsible for evolving the system and registering the particles at specific locations. Here, without the loss of generality, the basic components of passive linear optical setups can be considered as single- and two-mode gates, such as mirrors, phase shifters, beam splitters and number-resolving detectors²⁷.

The spatio-temporal arrangement of optical designs has implications for the possible interpretations of the experiment. Namely, it is common to think that the flow of information (or propagation of causes) follows the pattern of connections in the circuit. That is, its physical carriers propagate along the paths in the circuit (whether they are particles, fields or hidden variables), and the information is processed in a modular manner. This view stems from the principle of local causality, which is especially appealing for space-like separated arrangements. See Fig. 1. It was Bell's fundamental insight that quantum correlations can be at odds with the causal structure implied by the design of an experiment¹⁻⁴. In the optical framework, this idea can be expressed in the following manner.

Definition 2. An optical experiment demonstrates non-local behaviour (or Bell non-locality) if the observed correlations cannot be explained with the variables tracing the pattern of connections in the experimental setup.

In other words, there is no local hidden variable model explaining the observed experimental statistics, where locality requires that the variables propagate from one optical element to another along the paths determined by the circuit. In practice, demonstrating non-locality in a given experiment boils down to a violation of some Bell-type inequality derived from the corresponding causal structure. For example, in Fig. 1, one interprets the state $|\psi\rangle$ together with the optical setup $\mathbb{U}$ as a source of particles that are sent to distant locations A, B and C where a Bell test can be performed²⁹. The opposite statement, regarding the apparent lack of non-locality, can only be justified by constructing an explicit local hidden variable model that is compatible with the experimental setup³⁰.

Main result

An optical experiment can be seen as a test of certain properties of the state that is supplied to the circuit. In **Definition 2**, the question of non-locality is posed for a given experimental setup, which includes both the state and the optical design. In this paper, we are interested in the property of the state *itself* without restricting it to a particular implementation. We ask whether the state can demonstrate Bell non-locality by admitting a wider range of possible arrangements. This idea comes from thinking about the state as a resource of non-locality in a larger class of experimental designs. Here we focus on the use of passive linear optics, as defined below.

Definition 3. A state $|\psi\rangle$ is a genuine non-local resource with respect to passive linear optics if it is capable of manifesting non-local behaviour in some passive linear optical experiment.

Note that to demonstrate non-locality of a given state, it is sufficient to provide a single example of a passive linear optical circuit in which Bell non-locality can be observed. However, showing the opposite for a given state requires constructing a generic local hidden variable model explaining correlations obtained in every conceivable experiment with passive linear optical setups.

Our main result takes the form of a simple criterion for states with a definite number of particles in Eq. (1)/(2).

Theorem 1. A state $|\psi\rangle$ is a non-local resource with respect to passive linear optics if, and only if, it is not of the single-mode type.

This provides a straightforward classification of multi-particle states based on a simple condition **Definition 1** and easy-to-understand interpretation. Interestingly, the only states that lack the non-local potential are those originating in the same mode. This leads to the following characteristics of states for both boson and fermion statistics; see illustration in Fig. 2.

Corollary 1. Neither fermions nor bosons provide a non-local resource for any single-particle state ($N = 1$). For the multi-particle case ($N \geq 2$), all fermion states are non-local resources, while some boson states are not (i.e., those of a single-mode type).

We refer to the **Methods** section for the detailed proof of **Theorem 1**. Let us conclude by unfolding a partial result which facilitates the full proof.

Yurke-Stoler test: A useful lemma

The key tool in our discussion of non-locality in the interferometric setups is the co-called *Yurke-Stoler test*. It is designed for the special case of two particles ($N = 2$) and two modes ($M = 2$), and asks about Bell non-locality of the following general state

$$|\phi\rangle = \left(\alpha \frac{a_1^{\dagger 2}}{\sqrt{2!}} + \beta a_1^\dagger a_2^\dagger + \gamma \frac{a_2^{\dagger 2}}{\sqrt{2!}} \right) |0\rangle, \quad (5)$$

where $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$ in the boson case, whilst for fermions we have $|\beta|^2 = 1$, $\alpha = \gamma = 0$. In the following, we show how the seminal idea in ref. 31 can be refined to serve the purpose at hand.

Yurke-Stoler test checks for a violation of Bell inequalities in the design depicted in Fig. 3. It takes the state Eq. (5) in two input modes 1 and 2, that are split into two dual-rail qubits $\{1,1'\}$ and $\{2,2'\}$. Then the modes $1'$ and $2'$ get swapped, and the dual-rail qubits are sent out to Alice and Bob, who perform a typical Bell experiment. We note that the choice of settings on Alice and Bob's side is unrestricted since any projective measurement on their dual-rail qubits can be implemented with passive linear optics. The protocol involves *post-selection*, which retains only those experimental runs

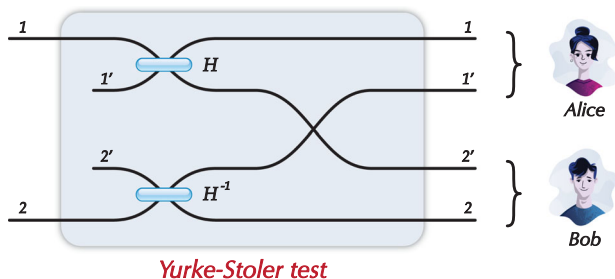


Fig. 3 | Yurke-Stoler test. A two-particle state, Eq. (5), in modes 1 & 2 is split by two beam splitters (Hadamard gates) into two dual-rail qubits $\{1,1'\}$ and $\{2,2'\}$. Then the modes $1'$ and $2'$ get swapped and Alice & Bob make the usual Bell test on their (dual-rail) qubits when each one of them gets a single particle.

in which a single particle is detected on Alice and Bob's side (i.e. when the dual-rail qubits are well-defined). Notably, the specifics of post-selection used in the experiment do not compromise the conclusions from the violation of Bell inequalities when the number of particles is conserved, as shown in refs. 32,33. It means that the protocol provides a genuine test of non-local correlations in the system.

This scheme was originally proposed by Yurke and Stoler to prove Bell non-locality for two particles coming from independent sources³¹. For a recent discussion see ref. 34. Here, we build on this idea taking it as a test of Bell non-locality for a general two-particle state $|\phi\rangle$ in Eq. (5), which via the setup in Fig. 3 evolves as follows

$$\begin{aligned} |\phi\rangle &\xrightarrow{H,H^{-1}} \left(\frac{\alpha}{2} \frac{(a_1^\dagger + a_{1'}^\dagger)^2}{\sqrt{2!}} + \frac{\beta}{2} (a_1^\dagger + a_{1'}^\dagger)(a_2^\dagger + a_{2'}^\dagger) + \frac{\gamma}{2} \frac{(a_2^\dagger + a_{2'}^\dagger)^2}{\sqrt{2!}} \right) |0\rangle \\ &\xrightarrow{1' \leftrightarrow 2'} \left(\frac{\alpha}{2} \frac{(a_1^\dagger + a_{2'}^\dagger)^2}{\sqrt{2!}} + \frac{\beta}{2} (a_1^\dagger + a_{2'}^\dagger)(a_2^\dagger + a_{1'}^\dagger) + \frac{\gamma}{2} \frac{(a_2^\dagger + a_{1'}^\dagger)^2}{\sqrt{2!}} \right) |0\rangle \\ &\xrightarrow{\text{post-select}} \left(\frac{\alpha}{\sqrt{2!}} a_1^\dagger a_{2'}^\dagger + \frac{\beta}{2} a_1^\dagger a_2^\dagger \pm \frac{\beta}{2} a_{1'}^\dagger a_{2'}^\dagger + \frac{\gamma}{\sqrt{2!}} a_{1'}^\dagger a_2^\dagger \right) |0\rangle, \quad (6) \end{aligned}$$

where the “ $-$ ” sign in “ $\pm$ ” refers to fermions that anti-commute (recall in that case we also have $|\beta|^2 = 1$, $\alpha = \gamma = 0$). Note that in the last line only the terms with a single particle in each dual-rail qubit $\{1,1'\}$ and $\{2,2'\}$ are retained. Clearly, this state is unnormalised due to post-selection, which succeeds with probability $1/2$. As a result, Alice and Bob share a dual-rail encoded two-qubit state in the form

$$|\chi\rangle \sim \frac{\alpha}{\sqrt{2!}} |\uparrow\uparrow\rangle + \frac{\beta}{2} |\uparrow\downarrow\rangle \pm \frac{\beta}{2} |\downarrow\uparrow\rangle + \frac{\gamma}{\sqrt{2!}} |\downarrow\downarrow\rangle, \quad (7)$$

where $|\uparrow\rangle \equiv |10\rangle$ and $|\downarrow\rangle \equiv |01\rangle$ is the computational basis for the respective dual-rail qubit $\{1,1'\}$ and $\{2,2'\}$.

Alice and Bob then conduct the standard Bell test with an arbitrary choice of measurements on either side. It will only reveal non-local correlations for some choice of settings if, and only if, $|\chi\rangle$ is an entangled state¹⁰. Conversely, the lack of non-local correlations observed in the Yurke-Stoler test requires that it must be a product state, i.e. we need to have $\beta^2 = 2\alpha\gamma$ ³⁵. Note that the latter condition implies that Eq. (5) simplifies to the single-mode type form (possible only for bosons)

$$|\phi\rangle = \frac{1}{\sqrt{2!}} \left(\sqrt{\alpha} a_1^\dagger + \sqrt{\gamma} a_2^\dagger \right)^2 |0\rangle = \frac{a_1^{\dagger 2}}{\sqrt{2!}} |0\rangle. \quad (8)$$

This proves a helpful result that can be used to analyse more complex interferometric designs.

Lemma 1. The state $|\phi\rangle$ in Eq. (5) does not manifest non-local correlations in the Yurke-Stoler test if, and only if, it is of a single-mode type, i.e., when we have $\beta^2 = 2\alpha\gamma$.

Note that for fermions, the Yurke-Stoler test is enough to demonstrate Bell non-locality for every two-particle ($N = 2$) and two-mode ($M = 2$) state. However, for the boson case, this setup is insufficient to prove non-locality for the single-mode type states. It will turn out that those states are locally simulable for any passive linear optical circuit. See **Methods** for the generic construction.

Discussion

The main interest of this work is to explore the boundary between the classical and quantum features of identical particles. We approach this task by drawing a line between the classical and quantum parts of experimental setups. This allows us to ask the question whether the state of the quantum part introduces a new quality that cannot be explained by classical means. In

other words, can the quantum state *itself* be thought of as a genuine non-local resource? For this reason, in the classical part we include only standard optical means of processing the state, consisting of passive linear optics and single-mode detection. Note that if any additional quantum components were allowed, then the answer would refer to both the state *and* the extra quantum resources used in the experiment. However, this is a very *different* question. Therefore, seeking an unambiguous answer, we avoid using any additional resources, such as extra quantum particles or ancillary states. In particular, this excludes POVMs or homodyne detection from consideration—if these were included, the conclusion of non-classicality would refer jointly to the state *and* the auxiliary resource used in the experiment, cf. refs. 36–41. In short, we stand on the position that adding an ancilla introduces extra quantum particles being a potential source of non-local effects, which undermines the conclusion regarding the non-locality of the studied state *itself* in such extended experimental procedures. Therefore, we exclude them from considerations⁴².

Our result shows a fundamental difference between the system of particles that evolved from a single mode and those that evolved from separate modes. We coined the simple term *single-mode type* state, which turns out to fully characterise the class of states incapable of exhibiting non-local effects in any passive linear optical experiment. It was shown that, for all those states, there exists a generic local hidden variable model that explains the observed statistics. The constructed model can be viewed as the maximum achievable as regards local simulation of multi-particle quantum states in passive linear optics. On the other hand, we have also shown that for every *non-single-mode type* state, a passive linear optical experiment can be found that demonstrates Bell non-locality. As an aside, let us mention that the devised optical schemes can be unfolded in a non-trivial space-separated manner and, furthermore, fulfil the so-called ‘*no-touching*’ property; see refs. 31,43 and refs. 34,44 for a recent discussion. This completes the characterisation of non-local states with a *definite number* of identical particles, which boils down to a few easy-to-check conditions defining single-mode type states. We note that our result considerably extends the discussion in ref. 23, where only the single-particle case was discussed.

The derived classification of non-local resources applies to both bosons and fermions, as illustrated in Fig. 2. Interestingly, all multiparticle fermion states provide a non-local resource for passive linear optics, whereas for bosons, there is a non-trivial class of single-mode type states that lack this property. This difference is due to the Pauli exclusion principle, which for fermions rules out their single-mode origin. Notably, in the non-local sector, bosons admit a wider range of states which exhibit peculiar properties.

Note that the notion of non-classicality can have different meanings. Here, it is understood as the ability of a state to demonstrate non-local behaviour in passive optical setups. For a different computational perspective, take the boson sampling protocol that also works in optical designs⁴⁵. It requires states being a non-local resource according to our classification, but for fermions, the computational advantage is lost. Another example is the difference in performance of bosons and fermions in various universal quantum computation schemes, e.g. see the KLM scheme⁴⁶ with bosonic linear optics (for fermions it is not universal). Let it serve as an illustration of the intricate relationship and differences between the fundamental concept of Bell non-locality, complexity and quantum advantage in information processing with identical particles. Those aspects go beyond the scope of the present paper; see refs. 14,20–22 for a discussion of those topics.

We remark that our problem can also be phrased in the modern framework of resource theories, e.g. ref. 47. In that language, single-mode type states constitute free states, and passive linear optics with single-mode detection define the set of free transformations. Then, any other multiparticle state provides a non-local resource. Another interesting perspective is offered by the reference frames framework¹⁸. Notably, in our problem, we are handicapped by the lack of an external reference frame, since we limit ourselves to particle detection only (and the number of particles is preserved). From the reference frames viewpoint, this introduces certain constraints on observables that can be effectively measured (for example, see the discussion of non-locality of the superposition of a single particle in two

modes in ref. 18). Those constraints predicted within the reference frames framework, via so-called superselection rules, crucially depend on the details of the physical implementation of the experimental setup, like the type of particles used (e.g. massive/massless, bosonic/fermionic) or the characterization of sources employed in the experiment (e.g. phase stability of used lasers). However, our approach is different as we do not utilise the reference frame approach, aiming at implementation-neutral results.

Finally, two general comments should be made about the physical focus of this research problem. Firstly, this work is *not* about the property of *entanglement*, which is a well-studied theoretical concept for *distinguishable* systems that is well-defined within the mathematical formalism of quantum theory¹¹. It is about the property called *Bell non-locality*^{1–4}, which concerns causal explanations of correlations observed in certain experimental setups that are crucially related to their spatial design (here, considered to be composed of passive linear optics and single-mode detection). This attitude is motivated by the fact that the definition of entanglement for a system of indistinguishable particles is challenging due to the mode description in the second-quantised theory^{15–22}. See ref. 14 for a recent review of the competing approaches, none of which directly refers to the causal notion of Bell non-locality discussed in this paper. In this sense, we turn towards the observed correlations and their causal interpretation, rather than considering the mathematical aspects of quantum formalism or the performance of entanglement in various quantum information tasks. Secondly, we are interested in systems of identical particles (bosons or fermions) that are closer to physical realisations, since any implementation necessarily involves particles that are deemed fundamentally indistinguishable. Although it may seem like a step backwards compared to the modern information-theoretic accounts, we believe this is a step in the right direction, following the slogan “*information is physical*”. This line of research highlights the fundamental quantum principle of indistinguishability of particles, which is sometimes overlooked despite being a core tenet of quantum mechanics. As eloquently stated by C. W. Misner, K. S. Thorne & J. A. Wheeler⁴⁸: “*No acceptable explanation for the miraculous identity of particles of the same type has ever been put forward. That identity must be regarded, not as a triviality, but as a central mystery of physics.*”

Methods

It is convenient to rephrase **Theorem 1** in the equivalent contrapositive form. Using **Definition 3**, it unfolds as follows.

Theorem 1’. *A state $|\psi\rangle$ is locally simulable in every passive linear optical experiment if, and only if, it is of a single-mode type.*

This is equivalent to the following two assertions.

Claim 1. (Necessity). *A state which does not manifest non-local correlations in any passive linear optical experiment is necessarily of the single-mode type.*

Claim 2. (Sufficiency). *All single-mode type states are locally simulable in arbitrary passive linear optical experiments.*

We note that in ref. 23, it was demonstrated that the single-particle case ($N = 1$) holds true (in that case all states are of the single-mode type²⁷). See also the proof of **Claim 2** below, where a generic local model for all single-mode type states is given, which includes the single-particle case.

In the following, we will focus on the multi-particle case $N \geq 2$. We first prove the easier fermion case and then proceed to sketch the proof for bosons.

Fermion case

There are no multi-particle states of the single-mode type for fermions, due to the Pauli exclusion principle. Therefore, for the proof, it is sufficient to indicate, for each fermion state, an experiment that shows non-local correlations.

We consider a state in Eq. (1) with all $n_i = 0$, 1. Since $N \geq 2$, there must exist a pair of indices k, l such that $\psi_{n_1 \dots n_M} \neq 0$ for $n_k, n_l = 1$ and some choice of the remaining n_j ’s. Without loss of generality, we take $k = 1$, $l = 2$ and denote this particular choice of the remaining indices $\tilde{n}_j$ for $j \geq 3$. That is, we

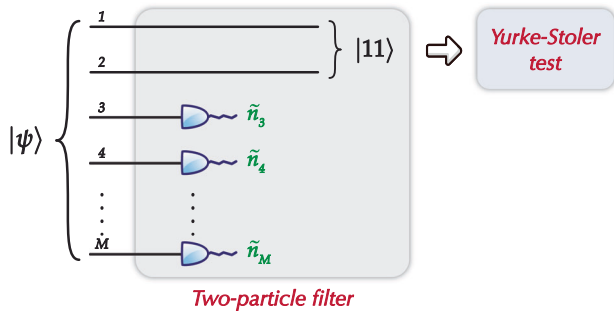


Fig. 4 | Filtering out two-particle state in the fermion case. Conditioning on the detection of $\tilde{n}_j$ particles in the array of detectors $j = 3, \dots, M$ prepares the event-ready state $|11\rangle$ in modes 1 & 2. Then the original Yurke-Stoler test can be made.

can assume that we have

$$\psi_{11\tilde{n}_3\dots\tilde{n}_M} \neq 0. \quad (9)$$

Now, consider the experimental procedure depicted in Fig. 4. This is an event-ready scheme in which the state $|11\rangle$ in modes 1 & 2 is heralded by the joint detection of $\tilde{n}_j$ particles in the respective modes $j = 3, \dots, M$. The condition Eq. (9) guarantees that this happens sometimes. Importantly, for fermions, there is only a single term with occupancy $\tilde{n}_3 \dots \tilde{n}_M$ in the expression in Eq. (1). The experiment ends with the Yurke-Stoler test carried out on such a prepared state in modes 1 & 2, which demonstrates non-local correlations; see Fig. 3 and Lemma 1.

Boson case

The boson statistics admits a wider variety of multi-particle states with more subtle correlation effects. This makes the above simple reasoning used for fermions not applicable in the boson case. In the following, we explain the main steps involved in the proof of Claim 1 and Claim 2. See Supplementary Information for the details.

It is crucial to observe that the coefficients in Eq. (1), for the single-mode type state can always be expressed in the product form⁴⁹

$$\psi_{n_1\dots n_M} = \left(\frac{N}{\vec{n}}\right)^{1/2} \prod_{i=1}^M U_i^{n_i}, \quad (10)$$

for some complex $U_1, \dots, U_M$ that normalise to one, i.e. $\sum_{i=1}^M |U_i|^2 = 1$. This is a simple consequence of the multinomial expansion applied to Definition 1 and Eq. (3).

Sketch of proof of Claim 1 (for bosons)

The idea of the proof is to consider examples of passive linear optical experiments and require that they *do not* exhibit non-local correlations. This will imply certain constraints on the coefficients in Eq. (1), necessitating the form of Eq. (10) which describes the *single-mode type* states. The key tool in the analysis is the Yurke-Stoler test and Lemma 1 discussed in the main text. It proves the case for two particles and two modes ($N=2, M=2$). In the following, this result will be used to derive constraints in certain experimental setups.

Many particles & two modes ($N \geq 2, M=2$)

A general two-mode state with arbitrary number of particles N can be written in form

$$|\phi\rangle = \sum_{n=0}^N \beta_n \frac{a_1^{\dagger n}}{\sqrt{n!}} \frac{a_2^{\dagger N-n}}{\sqrt{(N-n)!}} |0\rangle. \quad (11)$$

Let us start by considering a *two-particle filter*, shown in Fig. 5, which is designed to take an N particle state in the input and separate out a state with only two particles in the output. It splits each input mode 1 & 2 into $\{1, 1''\}$

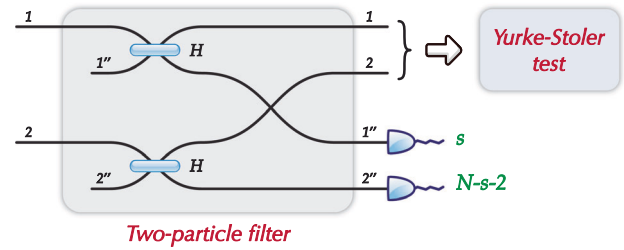


Fig. 5 | Yurke-Stoler type experiment with two-particle filter. In the initial phase $N-2$ particles are pulled out from the system by conditioning on s and $N-s-2$ particle detections in the auxiliary modes $1''$ & $2''$. For each $s = 0, 1, \dots, N-2$, this guarantees event-ready preparation of some two-particle state in modes 1 & 2 which then undergo the Yurke-Stoler test.

and $\{2, 2''\}$ respectively. Then, the number of particles is measured in the auxiliary modes, and only the cases with s particles in mode $1''$ and $N-s-2$ particles in mode $2''$ are retained. Such a defined post-selection, for each $s = 0, 1, \dots, N-2$ warrants that in modes 1 & 2 two particles remain. This filtering prepares some *event-ready* state on which the Yurke-Stoler to be performed. If no non-local correlations are to be observed with such a prepared state, then certain conditions must be met as explained in Lemma 1. Calculating explicitly, we get the following set of constraints

$$\beta_{s+1}^2 = \beta_s \cdot \beta_{s+2} \cdot \sqrt{\frac{s+2}{s+1}} \sqrt{\frac{N-s}{N-s-1}}, \quad (12)$$

for each $s = 0, 1, \dots, N-2$.

As it turns out, these conditions suffice to analyse all states of interest except for the so-called NOON states, which have the form

$$|\phi\rangle_{\text{NOON}} = \left(\beta_0 \frac{a_2^{\dagger N}}{\sqrt{N!}} + \beta_N \frac{a_1^{\dagger N}}{\sqrt{N!}} \right) |0\rangle, \quad (13)$$

i.e. when $\beta_1 = \dots = \beta_{N-1} = 0$ in Eq. (11). A quick inspection of Eqs. (12) shows that for the NOON states, these constraints are satisfied; hence, the *two-particle filter* in Fig. 5 cannot reveal non-local behaviour for that state. In this case, in order to get a non-trivial result we need to modify the optical setup by adding *quantum erasure*, as shown in Fig. 6. Here, the information about the number of particles in the respective modes $1''$ and $2''$ gets erased when the Hadamard transformation precedes the detection. This is a straightforward generalisation of the erasure of the which-path information for a single photon⁵⁰. For our purpose, it is sufficient to restrict the attention to the post-selected system with $N-2$ particles in mode $1''$ and 0 particles in mode $2''$. Then the resulting *event-ready* state in modes 1 and 2 is subject to the Yurke-Stoler test (We note that event-ready post-selection can be safely used in the arguments for non-locality⁵¹. A quick justification comes from the causal diagram analysis and the *d-separation rules*, à la ref. 33). Explicit calculation shows that the procedure does not reveal non-local correlations only if $\beta_0 = 0$ or $\beta_N = 0$, which follows from Lemma 1. That is, we have

$$\beta_1 = \dots = \beta_{N-1} = 0 \implies \beta_0 = 0 \quad \text{or} \quad \beta_N = 0, \quad (14)$$

meaning that every proper NOON state (for $\beta_0, \beta_N \neq 0$) is Bell non-local.

We have thus obtained enough constraints to provide the following technical result.

Lemma 2. The resolution of conditions in Eq. (12) and (14) implies that $\beta_n = \binom{N}{n}^{1/2} U_1^n U_2^{N-n}$ for some $|U_1|^2 + |U_2|^2 = 1$.

By comparing with Eq. (10), we conclude that any state $|\phi\rangle$ in Eq. (5) which *does not* show non-local correlations in the experimental setups in Figs. 5 and 6 is necessarily of a single-mode type. This proves Claim 1 for the case $N \geq 2$ and $M = 2$.

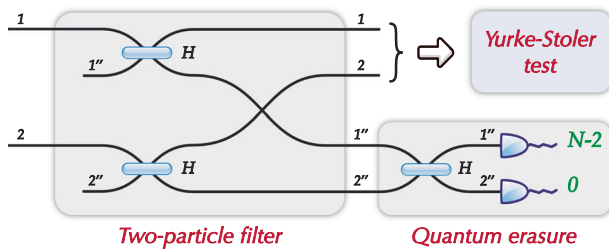


Fig. 6 | Yurke-Stoler type experiment with quantum erasure. Modification of the setup in Fig. 5 which erases the information about the number of particles in modes 1'' & 2''.

Many particles & many modes ($N \geq 2$, $M \geq 2$)

This is the most general case covered by **Claim 1**. To prove this result, we proceed by induction on the number of modes $M = 2, 3, \dots$ (for any number of particles $N \geq 2$). That is, we consider the following series of statements:

1. **Claim 1** holds for $M = 2$ (and any $N \geq 2$).
2. **Claim 1** holds for $M = 3$ (and any $N \geq 2$).
3. **Claim 1** holds for $M = 4$ (and any $N \geq 2$).

By mathematical induction, it is sufficient to justify:

- a. **Base case** ($M = 2$): **Claim 1** holds for $M = 2$ (and any $N \geq 2$).
- b. **Inductive step** ($M - 1 \rightsquigarrow M$): If **Claim 1** holds for $M - 1$ (and any $N \geq 2$), then it also holds for M (and any $N \geq 2$), for all $M \geq 3$.

Note that the **base case** was proved above; see **Many particles & two modes** ($N \geq 2$, $M = 2$). For the proof of the inductive step, we refer to **Supplementary Information**.

Sketch of proof of Claim 2 (for bosons)

We aim to construct a generic local hidden variable model that is capable of simulating any passive linear optical experiment with single-mode type states. This will require defining the *ontology* of what propagates in the circuit and how the *evolution* is implemented by the gates. Crucially, the propagation of variables in the model must follow the local pattern of interconnections in any given experimental design, in order to ensure the universality of the construction.

Without loss of generality, it suffices to consider two-mode gates (*beam splitters*) and single-mode gates (*phase shifters* and *detectors*). These basic elements wired into a complex circuit construct any passive linear optical experiment²⁷. Let us briefly recall how those components are described in quantum theory. A *beam splitter* is associated a the 2×2 unitary transformation $\mathbb{V}$ which acts on the intersecting paths s and t as follows

$$\begin{pmatrix} a_s^\dagger \\ a_t^\dagger \end{pmatrix} \longrightarrow \begin{pmatrix} a_s'^\dagger \\ a_t'^\dagger \end{pmatrix} = \mathbb{V} \begin{pmatrix} a_s^\dagger \\ a_t^\dagger \end{pmatrix}. \quad (15)$$

A *phase shifter* in path s introduces a phase $e^{i\varphi}$ where it is located

$$a_s^\dagger \longrightarrow a_s'^\dagger = e^{i\varphi} a_s^\dagger. \quad (16)$$

Finally, there are *single-mode detectors* which terminate evolution in a given path and provide classical information about the number of particles in that path. In our discussion of passive linear optical setups, their action can be effectively pulled out to the end of the experiment since all paths are eventually measured. Accordingly, an array of detectors that measure the state in Eq. (1) will register $n_1, n_2, \dots, n_M$ particles in the respective modes with the following statistics

$$|\psi\rangle \implies \text{Prob}_{|\psi\rangle}(n_1, \dots, n_M) = |\psi_{n_1 \dots n_M}|^2. \quad (17)$$

This completes the list of basic elements of passive linear optics and their quantum-mechanical description. Notably, for a general multi-particle

state, those rules lead to non-local correlations. In the following, we construct a generic local hidden variable model for the restricted class of single-mode type states. See Fig. 7 for illustration.

Construction of a generic local model

We will crucially exploit the property of single-mode type states, which allow for a compact representation

$$|\psi\rangle \xleftrightarrow{1:1} \vec{\alpha}_\psi. \quad (18)$$

It is defined through Eq. (10) where $\vec{\alpha}_\psi \equiv (U_1, \dots, U_M)$. We note that this property is specific to the single-mode type states since, in general, the coefficients $\psi_{n_1 \dots n_M}$ in Eq. (1) cannot be compactly encoded in only M numbers.

Lemma 3. For each single-mode type state $|\psi\rangle$, the associated vector $\vec{\alpha}_\psi$ is uniquely defined (up to the overall phase).

Let us define the *ontology* of the model by postulating that each mode $i = 1, \dots, M$ carries a complex number α_i and an integer k_i . The α_i 's will be called *amplitudes*, and k_i 's will be identified with the *number of particles* in i -th mode i . The *hidden variable space* is thus defined as

$$\Lambda := (\mathbb{C} \times \mathbb{N})^M = \mathbb{C}^M \times \mathbb{N}^M \ni (\vec{\alpha}, \vec{k}). \quad (19)$$

It is composed of subspaces $\Lambda_i := \mathbb{C} \times \mathbb{N}$ associated with each individual local mode, i.e. $\Lambda = \Lambda_1 \times \dots \times \Lambda_M$. In the following we will use the notation $\lambda \equiv (\vec{\alpha}, \vec{k}) \in \Lambda$.

Furthermore, to complete the model, we need to define how the ontic states $(\vec{\alpha}, \vec{k}) \longrightarrow (\vec{\alpha}', \vec{k}')$ evolve under the action of the basic elements building passive linear optical setups. Importantly, the *locality* condition requires that the gates defined below involve only the variables propagating along the paths meeting in the gate.

For the *beam splitter* associated with the unitary $\mathbb{V}$ in paths s and t , we postulate

$$\begin{aligned} (\alpha_s, \alpha_t) &\longrightarrow (\alpha'_s, \alpha'_t) := (\alpha_s, \alpha_t) \mathbb{V}, \\ (k_s, k_t) &\longrightarrow (k'_s, k'_t) \quad \text{with prob.} \begin{pmatrix} k \\ k'_s, k'_t \end{pmatrix} \frac{|\alpha'_s|^{2k'_s} |\alpha'_t|^{2k'_t}}{(|\alpha'_s|^2 + |\alpha'_t|^2)^k}, \end{aligned} \quad (20)$$

where $k := k_s + k_t = k'_s + k'_t =: k'$ is the number of particles, which is preserved. After the gate, the incoming particles in modes s and t are multinomially distributed between modes s and t with probabilities $p_s = |\alpha'_s|^2 / (|\alpha'_s|^2 + |\alpha'_t|^2)$ and $p_t = |\alpha'_t|^2 / (|\alpha'_s|^2 + |\alpha'_t|^2)$. Note that the latter are well-normalised and given by the transformed amplitudes (α'_s, α'_t) . Such a defined beam splitter is a local stochastic gate that operates exclusively in the intersecting paths.

The *phase shifter* in path s introduces phase $e^{i\varphi}$ to the amplitude in that path, i.e. we have

$$\alpha_s \longrightarrow \alpha'_s := e^{i\varphi} \alpha_s \quad \text{and} \quad k_s \longrightarrow k'_s := k_s, \quad (21)$$

which is a local deterministic gate.

Finally, each path ends with the *number-resolving detector*, which reveals the number of particles

$$(\alpha_s, k_s) \implies k_s. \quad (22)$$

Its action is deterministic and local.

This completes the list of basic elements in the model that enable the construction of any passive optical setup. It remains to demonstrate that the model reconstructs all the quantum-mechanical predictions for single-mode type states in any passive linear optical experiment.

Check of quantum-like behaviour

Let us postulate the *epistemic state* $\mu_\psi(\lambda)$ associated with the single-mode

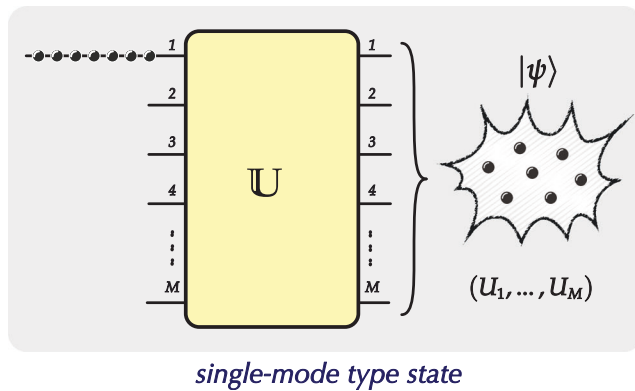
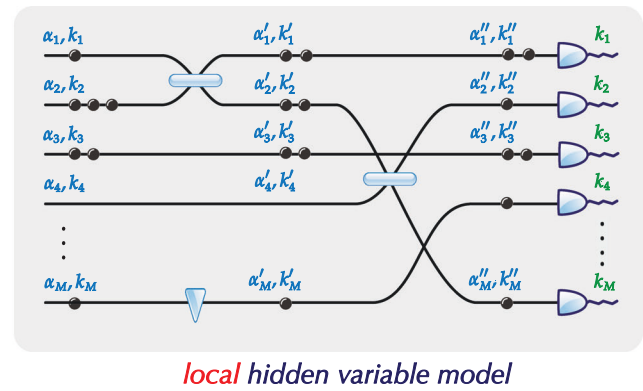


Fig. 7 | Single-mode type states and their local hidden variable model. On the left, *single-mode type states* are defined as reducible to a single mode by passive linear optical transformations. Hence, they admit a compact representation as a vector of complex amplitudes with only M components $(U_1, \dots, U_M)$, as defined in Eq. (10)/(18). It turns out that this vector is unique (up to the overall phase) and evolves locally under the action of the optical gates. On the right, the ontology of the hidden variable model is taken to consist of the *amplitude vector* $\vec{\alpha} = (\alpha_1, \dots, \alpha_M)$ together with the integer



vector $\vec{k} = (k_1, \dots, k_M)$ which describes the number of particles in each mode. Then, it is possible to define the local action of the gates (beam splitters, phase shifters and detectors) as given in Eqs. (20), (21) and (22), which will preserve the multinomial distribution of particles propagating in the system. This is enough to prove that predictions of the hidden variable model reconstruct the quantum-mechanical predictions for *single-mode type states* in any passive linear optical circuit, see **Lemma 4**.

type quantum state $|\psi\rangle$. Using the representation in Eq. (19), we define

$$\mu_\psi(\lambda) \equiv \mu_\psi(\vec{\alpha}, \vec{k}) := \frac{1}{2\pi} \delta_{\vec{\alpha} \sim \vec{\alpha}_\psi} \left(\frac{N}{\vec{k}} \right) \prod_{i=1}^M |\alpha_i|^{2k_i}, \quad (23)$$

where $\delta_{\vec{\alpha} \sim \vec{\alpha}_\psi}$ is the Dirac δ -function equal 1 if $\vec{\alpha} = e^{i\varphi} \vec{\alpha}_\psi$ for some $\varphi \in [0, 2\pi)$, otherwise equal 0. In other words, the epistemic state $\mu_\psi(\lambda)$ is *uniformly* distributed on $\vec{\alpha}$'s equal up to overall phase to $\vec{\alpha}_\psi$, and $\vec{k}$'s are *multinomially* distributed with probabilities $p_i = |\alpha_{\psi_i}|^2$.

The crucial observation is that the above-defined class of epistemic states $\mu_\psi(\lambda)$ is closed under the dynamics generated by the gates in the model and the evolution is congruent with the quantum-mechanical description.

Lemma 4. (*Equivariance*). *The epistemic states in Eq. (23) are preserved under the action of any configuration of gates defined in the model. Furthermore, their evolution $\mu_\psi(\lambda) \rightarrow \mu_{\psi'}(\lambda)$ follows the quantum-mechanical rules $|\psi\rangle \rightarrow |\psi'\rangle$ as laid out in Eqs. (15) and (16), and the observed experimental statistics is given by Eq. (17).*

In conclusion, when restricted to the class of epistemic states $\mu_\psi(\lambda)$, the predictions of the model are identical to the quantum-mechanical description of single-mode type states. This is sufficient to justify the existence of a generic local hidden variable model that simulates the behaviour of any single-mode type state in arbitrary passive linear optical experiments. Notice the resemblance of our construction model to the de Broglie-Bohm (or pilot wave) theory⁵², with the caveat that the latter is non-local. Our model can be viewed as the maximum achievable as regards local simulation of multi-particle quantum states in passive linear optical setups.

Data availability

No datasets were generated or analysed during the current study.

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30. Note that, in this case, the pattern of interconnections in the optical circuit is important, since the spatial design can be arranged in a non-trivial way that determines how the variables are allowed to propagate through the circuit. Given the flexibility of possible spatial configurations, it is necessary to be able to explain the internal flow of variables within the circuit itself, cf. ref. 53. Ultimately, we will need a generic hidden variable model simulating given states in arbitrary optical designs, cf. ref. 23.
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Author contributions

Both authors carried out the research. P.B. wrote the paper and both authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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